TABLE OF CONTENTS

Abstract i

Acknowledgement ii

Table of Contents iii

List of Abbreviations ix

List of Figures xi

List of Tables xiv

Chapter 1 Introduction 1

1.1 Background of the Study 1

1.2 Malta (Citrus sinensis) and the Garhwal Region 3

1.3 Economic, Social and Cultural Importance of Malta in Uttarakhand 5

1.4 Distribution Channels for Malta: Current Patterns and Challenges 6

1.5 Policy Context: Horticulture Missions and the "Malta Mission" 9

1.6 Problem Statement 10

1.7 Research Questions 11

1.8 Objectives of the Study 12

1.9 Scope of the Study 13

1.10 Significance of the Study 13

1.11 Organisation of the Thesis 14

Chapter 2 Review of Literature 16

2.1 Introduction 16

2.2 Approach to the Review of Literature 17

2.3 Literature on Malta (Citrus sinensis), Citrus Crops, and Hill Horticulture 19

2.4 Literature on Agricultural Marketing and Distribution Channels 22

2.5 Literature on Marketing Efficiency in Fruits and Perishable Commodities 23

2.6 Literature on Post-Harvest Management, Quality, and Perishability 25

2.7 Literature on Supply Chain Optimization in Agri-Food Systems 26

2.8 Theoretical Review: The Resource-Based View (RBV) 27

2.9 Literature on Multi-Criteria Decision-Making in Agricultural and Supply Chain Contexts 29

2.10 Review of Studies Using Fuzzy AHP 31

2.11 Review of Studies Using Fuzzy ELECTRE 33

2.12 Synthesis of Literature and Thematic Insights......35
2.13 Research Gaps......36
2.14 Conceptual Orientation of the Present Study......37
2.15 Chapter Summary......38

Chapter 3 Research Objectives, Research Gaps and Problem Statement......40
3.1 Introduction......40
3.2 Contextual Background to the Research Problem......41
3.3 Synthesis of Key Insights from the Literature Review......43
3.4 Research Gaps......44
3.5 Statement of the Problem......46
3.6 Research Questions......49
3.7 Research Objectives......49
3.8 Justification of the Study......51
3.9 Conceptual Positioning of the Present Study......52
3.10 Chapter Summary......53

Chapter 4 Research Methodology......55
4.1 Introduction......55
4.2 Research Design......56
4.2.1 Nature and orientation of the study......56
4.2.2 Research approach......57
4.2.3 Appropriateness of the design for the Malta channel in Garhwal......58
4.2.4 Justification for a cross-sectional design......58
4.3 Theoretical Foundation......59
4.3.1 The Resource-Based View......59
4.3.2 The VRIO framework and the cost, time, and quality dimensions......60
4.4 Study Area......61
4.5 Unit of Analysis and Stakeholder Coverage......62
4.6 Data Sources......62
4.6.1 Primary data......62
4.6.2 Secondary data......63
4.7 Questionnaire Development......63
4.7.1 Adoption and modification of a validated instrument......63
4.7.2 The adaptation process......64
4.7.3 Content validity and expert review......66
4.7.4 Structure of the final instrument......66
4.8 Sampling Design......68

4.8.1 Identification of the study blocks......71
4.8.2 Rationale for the selection of the districts and blocks......72
4.9 Sample Size Justification......76
4.9.1 Evidence from the literature and a note of caution......77
4.10 Reliability and Consistency Testing......79
4.11 Refinement of Factors......81
4.11.1 Identification of factors......82
4.11.2 Screening, refinement, and elimination of factors......84
4.11.3 The final set of 82 retained factors......86
4.11.4 Descriptive characteristics and validity of the responses......94
4.12 Inclusion of Fuzzy AHP in the Research Methodology......96
4.12.1 Conceptual introduction......96
4.12.2 Building blocks: goal, criteria, alternatives, and the fuzzy scale......97
4.12.3 Step-by-step procedure......98
4.13 Inclusion of Fuzzy ELECTRE in the Research Methodology......100
4.13.1 Conceptual introduction......100
4.13.2 Step-by-step procedure......101
4.14 Integrated Analytical Workflow......103
4.15 Ethical Considerations......103
4.16 Chapter Summary......104

Chapter 5 Data Analysis......105
5.1 Introduction......105
5.2 Profile of Respondents......106
5.3 Reliability and Item-Level Diagnostics......108
5.4 Construction of Crisp and Fuzzy Decision Matrices......110
5.5 Fuzzy AHP Analytical Procedure......113
5.5.1 Rationale, goal, criteria, and alternatives......113
5.5.2 Pairwise comparison and consistency......113
5.5.3 Fuzzification, geometric means, and fuzzy weights......114
5.5.4 Derivation of the Cost weight......116
5.6 Fuzzy ELECTRE I Analytical Procedure......117
5.6.1 Rationale and normalisation......117
5.6.2 Concordance, discordance, and dominance......118
5.6.3 Net flows and final ranking......121
5.7 Sensitivity Analysis......124
5.8 Summary of Analytical Pipeline......126

Chapter 6 Results **128**
6.1 Introduction 128
6.2 Descriptive Results 130
6.3 Reliability and Consistency of Scales 132
6.4 Fuzzy AHP Results 134
6.4.1 Rationale, goal, criteria, and alternatives 134
6.4.2 Construction of the empirical priority signal 134
6.4.3 Pairwise comparison and the Saaty-rounded matrix 135
6.4.4 Fuzzification and Buckley's geometric-mean method 136
6.4.5 Row geometric means, fuzzy weights, and defuzzification 137
6.4.6 Consistency assessment 138
6.4.7 Interpretation of the criterion weights 139
6.5 Fuzzy ELECTRE I Results 141
6.5.1 Crisp and fuzzy decision matrices 141
6.5.2 Normalisation and weighted normalised matrix 143
6.5.3 Concordance and discordance analysis 144
6.5.4 Dominance and outranking 146
6.5.5 Net-flow ranking 147
6.5.6 Interpretation of the ranking 149
6.6 Stage Criticality Outcomes 149
6.6.1 Competitive profiles of the stages 151
6.6.2 Factor-level findings by dimension and attention priority 153
6.7 Integrated Interpretation of Fuzzy AHP and Fuzzy ELECTRE 156
6.8 Sensitivity Analysis and Robustness 159
6.9 Positioning of the Findings against Prior Literature 161
6.10 Policy and Managerial Implications 162
6.11 Transportation Model for Optimising Malta Distribution 164
6.11.1 Rationale and Model Formulation 165
6.11.2 Supply, Demand and Cost Data 165
6.11.3 Solution Procedure 166
6.11.4 Optimal Distribution Plan 166
6.11.5 Interpretation and Managerial Insights 168
6.12 A Combined Multi-Echelon Transportation Model for Malta under Uncertainty 169
6.12.1 Rationale and Network Structure 169
6.12.2 Assumptions 170
6.12.3 Notation: Sets, Parameters and Decision Variables 170

6.12.4 The Combined Transportation Model......171
6.12.5 Modelling Demand Uncertainty......172
6.12.6 Solution Approach and Relevance to Malta......173
6.13 Answers to the Research Objectives......174
6.14 Summary of Results......176

Chapter 7 Discussion......177
7.1 Introduction to the Discussion......177
7.2 Discussion of the Criterion Priorities......178
7.2.1 Why cost dominates......178
7.2.2 Why quality is a close second......178
7.2.3 Why time ranks third......179
7.3 Discussion of the Stage Criticality......179
7.3.1 The primacy of the farm......179
7.3.2 The structural weight of the mandi......179
7.3.3 Transport, trader, and retail......180
7.4 Discussion of Concentration and Robustness......180
7.5 Discussion through the Resource-Based View......180
7.6 Reconciliation of the Findings with Theory......182
7.7 Discussion in the Policy and Development Context......183
7.8 Concluding Reflections on the Findings......183

Chapter 8 Conclusion and Future Directions......185
8.1 Major Conclusions......185
8.1.1 Region-specific conclusions for the Garhwal districts......187
8.2 Contributions of the Study......189
8.2.1 Empirical contribution......189
8.2.2 Theoretical contribution......189
8.2.3 Methodological contribution......190
8.3 Policy Recommendations......190
8.3.1 Turning Malta into a regional opportunity: an agenda for the Government of Uttarakhand......192
8.4 Implications of the Study......193
8.4.1 Social implications......194
8.4.2 Managerial implications......194
8.4.3 Practical implications......194
8.5 Limitations of the Study......195
8.6 Directions for Future Research......196

References..199

LIST OF ABBREVIATIONS

Abbreviation	**Expansion**
AHP	Analytic Hierarchy Process
APEDA	Agricultural and Processed Food Products Export Development Authority
APMC	Agricultural Produce Market Committee
AVE	Average Variance Extracted
CI	Consistency Index
CR	Consistency Ratio
DEA	Data Envelopment Analysis
DEMATEL	Decision-Making Trial and Evaluation Laboratory
ELECTRE	ELimination Et Choix Traduisant la REalite (Elimination and Choice Expressing Reality)
FAO	Food and Agriculture Organization of the United Nations
FAPI	Factor Attention-Priority Index
FPO	Farmer Producer Organisation
GPS	Global Positioning System
ICAR	Indian Council of Agricultural Research
MABAC	Multi-Attributive Border Approximation Area Comparison
MARCOS	Measurement of Alternatives and Ranking according to Compromise Solution
MCDM	Multi-Criteria Decision-Making
MIDH	Mission for Integrated Development of Horticulture
MODI	Modified Distribution (method)
MSP	Minimum Support Price
MT	Metric Tonne
NAM	National Agriculture Market (e-NAM)
NH	National Highway
NHB	National Horticulture Board
NITI Aayog	National Institution for Transforming India
ODOP	One District One Product

PMFME	Pradhan Mantri Formalisation of Micro Food Processing Enterprises
RBV	Resource-Based View
RCI	Relative Criticality Index
RI	Random Index
SCM	Supply Chain Management
SHG	Self-Help Group
TFN	Triangular Fuzzy Number
TOPSIS	Technique for Order of Preference by Similarity to Ideal Solution
UUHF	Uttarakhand University of Horticulture and Forestry
VAM	Vogel's Approximation Method
VCSG	Veer Chandra Singh Garhwali
VIKOR	VIseKriterijumska Optimizacija I Kompromisno Resenje (Multi-Criteria Optimisation and Compromise Solution)
VRIO	Valuable, Rare, Inimitable, Organised

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1.1	Schematic Map of the Study region; the Garhwal districts of Uttarakhand covered in the study (Pauri Garhwal, Tehri Garhwal, Chamoli and Rudraprayag)	4
Figure 1.2	Schematic of the existing Malta distribution channels from farm to consumer (Channel I to Channel VI)	7
Figure 2.1	Conceptual map of the literature domains relevant to the study	19
Figure 2.2	Thematic framework linking the literature streams to the present study	31
Figure 2.3	Research gap framework and positioning of the present study	37
Figure 3.1	Conceptual pathway from the review of literature to the problem statement, research questions, and objectives	41
Figure 3.2	Problem framework for Malta distribution channel optimisation in Garhwal	48
Figure 3.3	Research logic map linking contextual, empirical, dimensional, theoretical, and methodological gaps to the study objectives	53
Figure 4.1	Roadmap of the research methodology chapter	56
Figure 4.2	Research design positioned on the research onion	57
Figure 4.3	Malta (Citrus sinensis) distribution channel	58
Figure 4.4	RBV-VRIO lens linking channel resources to the cost, time, and quality dimensions	60
Figure 4.5	Map of the study area (Teal Color) within the Garhwal region	61
Figure 4.6	Instrument adoption and adaptation process	65
Figure 4.7	Blueprint of the survey instrument (items per stage x dimension)	67
Figure 4.8	Multi-stage stratified sampling procedure	69
Figure 4.9	Stakeholder composition of the achieved sample	70
Figure 4.10	District-wise distribution of the sample	70
Figure 4.11	Demographic profile of respondents (age, education, experience)	71
Figure 4.12	Cronbach's alpha reliability across the fifteen sub-scales	80
Figure 4.13	Factor refinement funnel	81
Figure 4.14	Mean importance of cost, time, and quality across the five channel stages	82
Figure 4.14a	Factor identification and screening from 144 to 82 factors	84

Figure 4.14b Factor elimination decision logic.....85
Figure 4.15 Distribution of Likert responses by performance dimension.....94
Figure 4.16 Inter-correlation among the fifteen sub-scales.....95
Figure 4.17 Perceived importance profile across stages (radar view).....95
Figure 4.18 Decision hierarchy for the Fuzzy AHP model.....97
Figure 4.19 Triangular fuzzy numbers underlying the linguistic comparison scale.....98
Figure 4.20 Procedure of Fuzzy AHP.....99
Figure 4.21 Concordance and discordance logic in Fuzzy ELECTRE.....100
Figure 4.22 Procedure of Fuzzy ELECTRE I.....102
Figure 4.23 Integrated analytical workflow of the study.....103
Figure 5.1 District-wise distribution of the sample.....106
Figure 5.2 Stakeholder composition of the achieved sample.....107
Figure 5.3 Demographic profile of respondents (age, education, experience).....108
Figure 5.4 Cronbach's alpha across the fifteen sub-scales.....109
Figure 5.5 Crisp decision-matrix heatmap.....111
Figure 5.6 Radar plot of stage performance across the three criteria.....112
Figure 5.7 Fuzzy decision matrix visualised as (l, m, u) intervals.....113
Figure 5.8 Fuzzy AHP criterion weights with fuzzy spread.....115
Figure 5.9 Concordance index heatmap.....119
Figure 5.10 Discordance index heatmap.....120
Figure 5.11 Aggregate dominance heatmap.....121
Figure 5.12 Net concordance and net discordance flow by stage.....122
Figure 5.13 Final ranking of the five channel stages (Fuzzy ELECTRE I).....123
Figure 5.14 Sensitivity of the ranking to weight perturbation.....125
Figure 5.15 Analytical pipeline of Chapter 5.....126
Figure 6.1 Headline outcomes of the study at a glance.....129
Figure 6.2 District-wise distribution of the sample.....130
Figure 6.3 Stakeholder composition of the achieved sample.....132
Figure 6.4 Cronbach's alpha across the fifteen sub-scales.....133
Figure 6.5 Heatmap of the pairwise comparison matrix.....136
Figure 6.6 Triangular fuzzy numbers for the linguistic comparison scale.....137
Figure 6.7 Fuzzy AHP criterion weights with fuzzy spread (whiskers).....138

Figure 6.8 Relative weight of the three decision criteria....140
Figure 6.9 How cost, time and quality build each stage's composite priority....140
Figure 6.10 Crisp decision-matrix heatmap....142
Figure 6.11 Radar plot of stage performance across the three criteria....142
Figure 6.12 Weighted decision matrix (crisp score × criterion weight)....143
Figure 6.13 Concordance matrix heatmap....145
Figure 6.14 Discordance matrix heatmap....146
Figure 6.15 Aggregate dominance matrix heatmap....147
Figure 6.16 Final ranking of Malta channel stages (Fuzzy ELECTRE I)....148
Figure 6.17 Relative Criticality Index of the five channel stages....150
Figure 6.18 Concentration of channel criticality (Pareto view)....151
Figure 6.19 Competitive profile of each stage by criterion (1 = best, 5 = weakest)……152
Figure 6.20 Strategic priority map: criticality versus cost vulnerability....153
Figure 6.7a Fuzzy AHP decision hierarchy with global criterion weights....156
Figure 6.7b Fuzzy ELECTRE I global ranking of the five channel stages, with Relative Criticality Index and net outranking flow....157
Figure 6.21 Rank stability underweight perturbation....160
Figure 6.22 Four independent grounds for confidence in the outcomes....161
Figure 6.23 Intervention priority tiers derived from the criticality outcomes....164
Figure 6.24 Optimal Malta transportation network: district-to-market flows....167
Figure 6.25 Optimal dispatch plan by route (quintals and unit cost)....167
Figure 6.26 Total transportation cost: optimised plan versus naïve baseline....168
Figure 6.27 Structure of the four-echelon combined transportation network for Malta 186
Figure 8.1 Study's conclusions....186
Figure 8.3 Implementation roadmap (Gantt chart) for the prioritised Malta interventions, by tier and time horizon....191
Figure 8.2 Turning Malta into a regional opportunity for Uttarakhand....193

LIST OF TABLES

Table No.	Title	Page No.
Table 1.1	Area, Production and Productivity of Citrus (Including Malta) in Uttarakhand, Selected Years (2008–2009 to 2022–2023)	2
Table 1.3	Indicative Share of Marketed Malta Handled by Each Type of Distribution Channel and Approximate Producer's Share of Consumer Price	8
Table 1.2	Comparative Summary of Key Horticulture Missions in Uttarakhand (Apple, Kiwi and the Proposed Malta Mission)	10
Table 2.1	Summary of key studies on Malta (Citrus sinensis) in Uttarakhand and related hill regions	21
Table 2.2	Major studies on fruit and vegetable marketing efficiency	24
Table 2.3	Studies on post-harvest quality and perishability in horticultural supply chains	26
Table 2.4	Studies applying the Resource-Based View in agri-food or supply-chain research	29
Table 2.5	Review of studies applying Fuzzy AHP in agriculture, logistics, and supply chains	33
Table 2.6	Review of studies applying Fuzzy ELECTRE / outranking methods in agriculture, logistics, and supply chains	34
Table 2.7	Research gap matrix	36
Table 3.2	Research gap matrix	46
Table 3.3	Problem dimensions in Malta distribution channels	48
Table 3.4	Mapping of research questions to research objectives	50
Table 3.5	Justification matrix showing the academic, policy, social, and practical relevance of the study	52
Table 4.1	Rationale for selecting a cross-sectional over a longitudinal design	59
Table 4.2	District profile of the study area	62
Table 4.3	Questionnaire adaptation framework based on Negi (2018)	65
Table 4.4	Operational definition of the three constructs with illustrative items	68
Table 4.5	Stakeholder composition and district-wise sample allocation (N = 1,300)	71
Table 4.6	Selected demographic characteristics of respondents	71
Table 4.5a	Basis for the identification of study blocks, by district	72
Table 4.5b	District production metrics and horticultural profile	73

Table 4.5c Production characteristics and research comparative matrix........................74

Table 4.5d Integrated selection rationale, production profiles, and marketing dynamics for the study sites..75

Table 4.7 Positioning of the present study against illustrative methodological literature 78

Table 4.8 Reliability statistics of the fifteen sub-scales (N = 1,300)............................80

Table 4.9 Mean perceived importance of cost, time, and quality by channel stage (1–5 scale)...82

Table 4.9a Sources used for the identification of factors..83

Table 4.9b Thematic classification of the initial pool of 144 factors.............................83

Table 4.9c Criteria used for the screening and elimination of factors............................85

Table 4.9d Categories of eliminated factors and reasons for elimination.......................85

Table 4.9e Stage-wise reduction of the factor pool from 144 to 82...............................85

Table 4.9f Complete pool of 144 identified factors with retention status......................86

Table 4.9g The final set of 82 retained factors, by stage and dimension........................91

Table 4.10 Linguistic scale and corresponding triangular fuzzy numbers for Fuzzy AHP 97

Table 4.11 Stepwise summary of the Fuzzy ELECTRE I procedure...........................102

Table 5.1 Geographic distribution of the sample across districts...............................106

Table 5.2 Stakeholder-group composition of the sample...107

Table 5.3 Demographic profile of respondents (age and experience)........................108

Table 5.4 Reliability of the fifteen sub-scales (Jamovi)..109

Table 5.5 Crisp decision matrix (mean rating by stage and criterion)........................110

Table 5.6 Fuzzy decision matrix (triangular fuzzy numbers, l, m, u).........................112

Table 5.7 Crisp pairwise comparison matrix for the three criteria.............................114

Table 5.8 Fuzzified pairwise comparison matrix..114

Table 5.9 Row geometric means, fuzzy weights, and defuzzified criterion weights..115

Table 5.10 Step-by-step derivation of the Cost criterion weight (Buckley method)….117

Table 5.11 Concordance index matrix...118

Table 5.12 Discordance index matrix...119

Table 5.13 Aggregate dominance matrix (1 = row outranks column)..........................121

Table 5.14 Net concordance/discordance flows and final ranking..............................122

Table 5.15 Stage ranks underweight-perturbation scenarios.......................................124

Table 6.1 Headline outcomes of the integrated Fuzzy AHP–Fuzzy ELECTRE I analysis.....129
Table 6.2 Geographic distribution of respondents (N = 1,300).....130
Table 6.3 Stakeholder-group composition of the sample.....131
Table 6.4 Reliability of the fifteen sub-scales (Jamovi).....132
Table 6.5 Empirical priority signals for the three criteria.....134
Table 6.6 Saaty-rounded pairwise comparison matrix.....135
Table 6.7 Fuzzified pairwise comparison matrix.....137
Table 6.8 Row geometric means, fuzzy weights, and defuzzified criterion weights.....137
Table 6.9 Crisp decision matrix (stage means by criterion).....141
Table 6.10 Fuzzy decision matrix (triangular fuzzy numbers).....143
Table 6.11 Concordance matrix ($\bar{c}$ = 0.500).....144
Table 6.12 Discordance matrix ($\bar{d}$ = 0.591).....145
Table 6.13 Aggregate dominance matrix (1 = row outranks column).....146
Table 6.14 Net-flow ranking of the five channel stages.....148
Table 6.15 Relative Criticality Index and criticality concentration.....150
Table 6.16 Competitive profile of the five stages.....152
Table 6.16A Distribution of the 82 validated factors across the three performance dimensions.....153
Table 6.16B Factor attention-priority index by stage and dimension (% of total decision-criticality).....154
Table 6.16C Attention-priority tiers for the 82 factors.....154
Table 6.16D Factors requiring attention (Tiers 1 and 2), by priority tier and dimension 157
Table 16.17 Fuzzy AHP decision hierarchy and derived criterion weights.....157
Table 16.18 Fuzzy ELECTRE I global ranking.....158
Table 6.17 Weighted composite scores corroborating the outranking.....159
Table 6.18 Stage ranks underweight-perturbation scenarios.....159
Table 6.19 Robustness outcome summary.....161
Table 6.20 Positioning of the study's outcomes against prior literature.....162
Table 6.21 Outcome-to-implication mapping.....164
Table 6.22 Supply, demand and unit transportation-cost matrix (₹ per quintal).....166
Table 6.23 Optimal dispatch plan for Malta (quintals).....166
Table 6.24 Notation of the combined multi-echelon transportation model.....170

Table 7.1 Findings, interpretations, and their principal links to the literature............177
Table 8.1 Summary of the study's major conclusions...186
Table 8.1a Region-specific conclusions for the four study districts.............................188
Table 8.2 Prioritised policy and managerial recommendations..................................191
Table 8.3 A Government of Uttarakhand opportunity agenda for Malta....................192
Table 8.4 Social, managerial, and practical implications of the study........................195